Bitumen tests
- flash & fire pt test
  - kya testh ye
  - Apparatus
  - flash pt
    - Defn
    - temp
  - fire pt
    - Defn
- Solubility test
- Spot test
  - kiske liye hai

flash & fire pt test

(13.6)

- These are primarily safety tests.
- Pensky martin closed cup & open cup apparatus
- flash point:
  The lowest temperature @ which the vapour of a substance momentarily takes fire in form of flash
- 175°C
- fire pt: the lowest temp at which material gets ignited & burns under specified condition of test.

Solubility test

Spot test

- Test for detection of overheated or cracked bitumen

Bitumentests
- flash & firept test
  - kya testh ye
  - Apparatus
  - flashpt
    - Defn
    - temp
  - firept
    - Defn
- Solubilitytest
- Spot test
  - kiske liye hai

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Solubility test

Spot test

- Test for detection of overheated or cracked bitumen

13.2

Bitumen tests
- Spot test
  - kinematic
  - water content
    - Max wtr content in bitumen
  - float test
    - kya use
    - kaise
    - kya indicates
    - unit

naptha is used

Water content test

- Maximum water content in bitumen should not be less than 6.2% by weight

float test

- it is also a measure of consistency and viscocity.
- Higher the test value, stiffer
- od harder the material
- measured in time taken in seconds

Grading of bitumen

Grading of bitumen

- In india bitumen grade is represented by penetration value, penetration grade.
- Eg 80/100, 60/70, 30/40

Grading of bitumen

- New method

Now a more rational method of grading is used by BIS — in terms of viscosity grading. (VG)

	Absolute Viscosity	kinematic viscosity	Penetration val
VG10 - kiemat virco / Pentration val	800 poise	250 centistoke	80-100
VG20	1600	300	60-80

General application

- General application

- VG 10
- VG 20
- VG-30
- VG-40

- VG 10: (80-100) spraying, paving applicn in cold region
- VG 20 (60-80) → paving application in cold region @ high altitude & northern part of india
- VG 30 (50-70) → paving application most part of india
- VG-40 (40-60): used in high stressed area such as intersections toll plaza truck layby

13-9) Grading of bit
- test temp
  - abs visc
  - kinemat visc
- TFOT
  - kya kinds of test ye test?

absolute viscoid test temperature 60°C
kinematic viscocity test temper 135°C
penetrn test 25°C

Thin film oven test

• The different viscocity values are observed in two simmilar viscocity grades of bitumen which is found by this test.

PMB --- full form ---- PMB polymer modified bitumen
CRMB ---- CRMB (rumbrubber modified bitumen

• Khan use krenge PMB aur CRMB ko These are used only in wearing coorse depends on requirement

Class 14 — 1 Aug 2024 (14.1)

Rigid pavement

- kya kya cheez hoti h is me?

Rigid pavement

- Consists of -
  - cement concrete
  - Wearing course

& it is of 3 layer system.

- kitni layers h?
  - Subgrade
  - base
  - Surface course

Detailing worksheet me milega)

- Pavement failure

- kya kya failures h?

Pavement failure.

- Rutting
- Revelling
- Pot holes

} CC + damar road

- Craziness
  - Aligator crack or map cracking -

- Aligator crack kisme dekhte h?

Seen in CC pavement

14.2 Rig pvmt Design steps

- load trfer
  - kis check
  - kc kerean
  - load carry rpad. kota h...
- formula
  - P kg-h...
  - S kg-h...
  - k kg/h...
- Radical rel S stiffness
  - kisse ati
  - rad of rel stiff ness

Rigid pavement design steps

1. load transfer

load carrying capacity is mainly due to rigidity & high modulus of elasticity of the slab itself

$E_c = 5000\sqrt{f_{ck}}$

$k = \frac{P}{\delta}$ $\quad \boxed{P = k\delta}$

Pressure = Mod of R x deflec

P is pressure below plate

δ is deflection at the center of plate

k is modulus of subgrade rxn.

2 Radius of relative stiffness

A certain degree of resistance to slab deflection is offered by the subgrade hence relative stiffness l_r is defined by westergaard for detailing deformn Characteristics

Design steps Rigid pvmt

2. Radius of relative stiffness (otd.

- formula.

$$\ell = \left[\frac{E_c h^3}{12k(1-\mu^2)}\right]^{1/4}$$

$E_c = 5000\sqrt{f_{ck}}$

$$\text{Radius of Relative stiffness} = \left[\frac{\text{Modulus of elasticity of concrete} \times (\text{slab thickness})^3}{12 \times \text{Modulus of sub grade reaction} \times (1 - \text{poissons ratio of conc (0.15)})^2}\right]^{1/4}$$

- units kg/cm²

E_c is in kg/cm²
h is in cm
k is in kg/cm³

} ℓ is obtained in cm.

- E
- h
- k
- ℓ

3. Equivalent rad of resisting secn

- formula when a < 1.724h

3. Equivalent radius of resisting section (b)

(i)
$$b = \left[\sqrt{1.6a^2 + h^2} - 0.675h\right]$$

[when $a < 1.724h$

- formula when a is different.

(ii) $b = a$ when (i) is not satisfied

(Generally yahi use hota h)

- a kya h?

'a' is radius of wheel load distrib?

- formula

$$a_{cm} = \sqrt{\frac{P\ \text{kg/cm}^2}{q_{cm}\pi}}$$

- p kya h

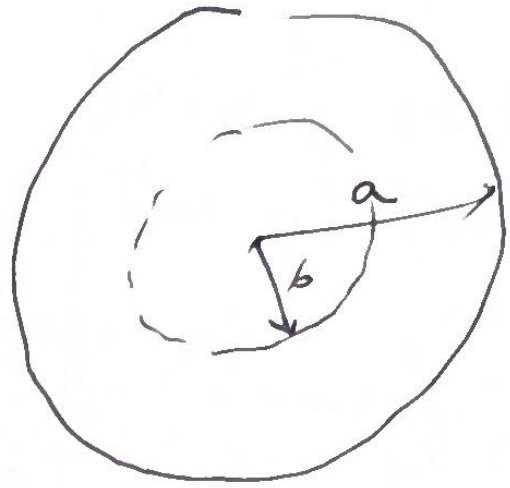
- q kya h?

q is contact pressure
p is pressure below the plate.

14.1) Compute the equivalent radius of resisting section of 15 cm slab if the ratio of radius of wheel distribution to thickness of slab is 0.5.

Ans:

b: ?



$a/h = 0.5$

$h = 15\,cm$

$a = 0.5 \times 15 = 7.5\,cm$

$1.724h = 1.724 \times 15 = 25.86\,cm.$

condition check

$a < 1.724h$

$7.5\,cm < 25.86\,cm$

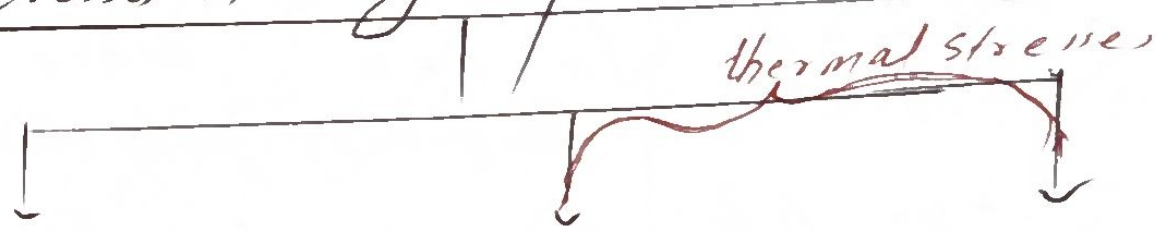
$b = \sqrt{1.6a^2 + h^2} - 0.675h$

$= \sqrt{1.6 \times 7.5^2 + 15^2} - (0.675 \times 15)$

$= 7.62\,cm$

Design of rig pvmt
- Stresses in pvmt

Stresses in rigid pavement



load stress (westergaards eqn for wheel load)

Warping stress (Bradbury)

frictional stress

kya kya stresses h usm aur kaunsa thermal h.

Design of rig pvmt - critical regions for rigid pavement are?

for rigid pavement critical regions are -

1. corner
2. edge
3. interior

- Westergaard modified westergaard smethy. farthaz

In westergaards theory load stresses are given for interior, edge, corner region.

In modified westergaards the load stresses are given for edge and corner region only.

- formula for corner loading

Corner loading stress $\sigma_c = \frac{3P}{kg/cm^2\ h^2}\left[1-\left(\frac{h\sqrt{2}}{l}\right)^{3.2}\right]$

14.6

Stresses in pvmt

- thermal stress
  - Warping stress formula @
    - (i) interior
      - α
      - C_x, C_y
      - x
      - t
    - (ii) edge
      - formula

Thermal stress

Warping stress (bradbury)

(i) interior

$$\sigma_w = \frac{E_c \alpha t}{2}\left[\frac{C_x + \mu C_y}{1-\mu^2}\right]$$

α is coefficient of thermal expansion per degree centigrade. $/^\circ C$

C_x & C_y are coefficients in x & y direction.

x is shorter dimension y longer dim

t is temp diff during day 6^{th} top & bottom layer of slab. $/in^\circ$.

(ii) edge

$$\sigma_e = \frac{C_x E_c \alpha t}{2}$$

$$\frac{C_y E_c \alpha t}{2}$$

} more

thermal stress important
Warping stress
@ (iii) corner
formula

$$\sigma_{wc} = \frac{E\alpha t}{3(1-\mu)}\sqrt{\left(\frac{a}{\ell}\right)}$$

frictional stress
frictional stress is always zero @ corner kyonki frictional stress ke kaaran slab ka soil se corner pe contact khatam ho jatah/ aur corner lift ho jatah jisse corner pe frictional stress dere top nahi hota.

frictional stress induced in bottom fiber due to contact with soil/subgrade should be equal to force in cc pavement due to movement. (for $\rightleftharpoons$)

force in cc pavement = frictional resistance due to subgrade

- frictional stress
- corner pe kitna h?
- kyon corner pe wob hota h jo hota h?
- $\rightleftharpoons$ ke liye kya = hone chahiye
- rig pvmt me friction ke respect me.

- 14.8 Thermal stress in rigpvmt
  - frictstress
    - formula for =
      - 2400 kg / m^3
      - kya h2.

$$f(B*h) = [B \times h \times \frac{L}{2}]\, rc \cdot f$$

$$f = \frac{Sc\, f\, L}{2}$$

$\rightarrow$ 2400 kg/m^3

kg/m^2

unit stress or frictional stress developed in cement concrete

$$= \frac{\text{mass density} \times \text{friction coeff} (1.5) \times \text{Slab length cm}}{2}$$

Bis slab width cm

his slab thickness cm

Critical combinn of stresses.

Critical combination of stresses

1. Summer midday
2. Winter midday
3. Winter midnight

- kya kya combos h2.
  - Smday
    - Critical stress kahan h?

① Summer midday

critical stress is defined for edge region

Critical combinn of stresses:

- ① Summer midday
  - formula
- ② Winter midday
  - Crit region kalvh
  - formula
- ③ Winter midnight
  - Crit region kavnight
  - formula

$\sigma_{critical} = \sigma_{load} + \sigma_{warping} - \sigma_{friction}$

② Winter midday

Crit region edge region.

$\sigma_{critical} = \sigma_{load} + \sigma_{warping} + \sigma_{friction}$

③ Winter midnight.

for winter midnight critical combination of stress obtained @ corner region

$\sigma_{crit} = [\sigma_{load\ corner} + \sigma_{warping\ corner}]$

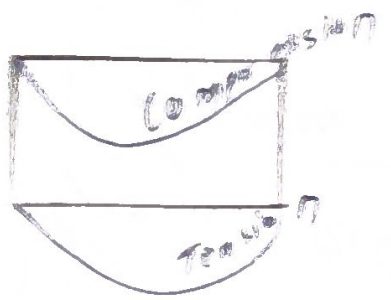
1-10

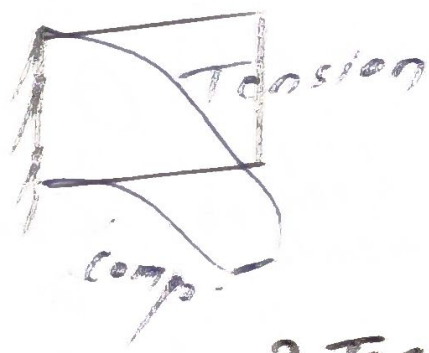
Stress pattern in load

- Stress
  - 1.1 distrib^n @ edge & interior
  - 1.2 distrib^n @ corner

Stress pattern compression or tension.

load stress





edge / interior] Compression (top) / Tension (bottom)

corner] Tension (top) / Compression (bottom)

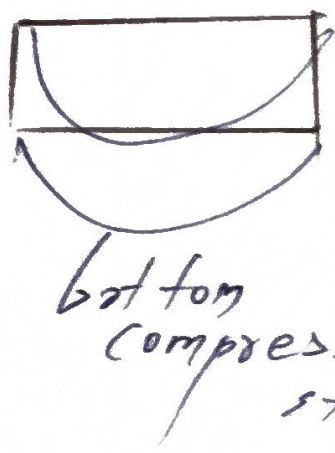
2 Warping stress.

- 2.1 Daytime me slab kaisa dikhta h? Top pe ky ata h?
- 2.2 Nightme kaisa dikhta h? kya Stress atah?

Warping Stress (same for edge corner & interior)

Due to warping stress @ daytime slab looks like →

@ nighttime →



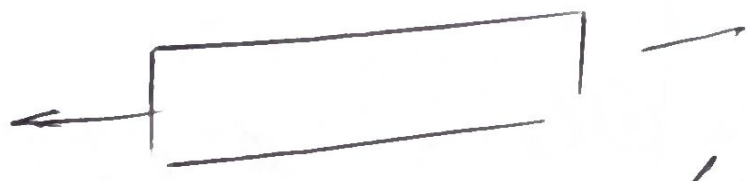
* Stress pattern
  - 3. frictional stress
    - 3.1 @ corner
    - 3.2 @ interior & edge
      - 3.2.1 summer
      - 3.2.2 Winter

frictional stress (only for edge interior, zero for corner)

(summer & winter)

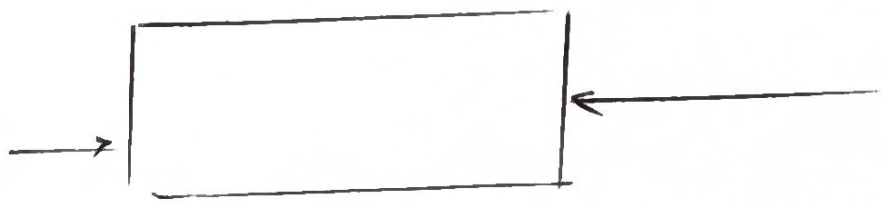
frictional stress is zero @ corner at interior and edge region

I. in summer



friction ke karan fail raha hai toh compressive stres develop hoga @ top & bottom.

II. In winter



Tensile stress develop hoga top & bottom me.

Design of tie bars
- provided where?

Design of tie bar - Provided across Longitudinal joint

Design of tie bars

As formula

Area of steel for tie bar

① $A_s = \frac{B \ell \rho_c}{\sigma_{st}}$

$\rho_c \rightarrow 2400\ kg/m^3$

$\sigma_{st} \rightarrow$ MPa (140 – 175 MPa)

mm^2/m length

σ_{st} is allowable working tensile stress used in steel in MPa

- Ld formula

② $L_d = \frac{\phi\, \sigma_{st}}{2 \tau_{bd}}$

ϕ is diameter of bar

τ_d is permissible bond stress of conc.

Class 15 22 Aug 24

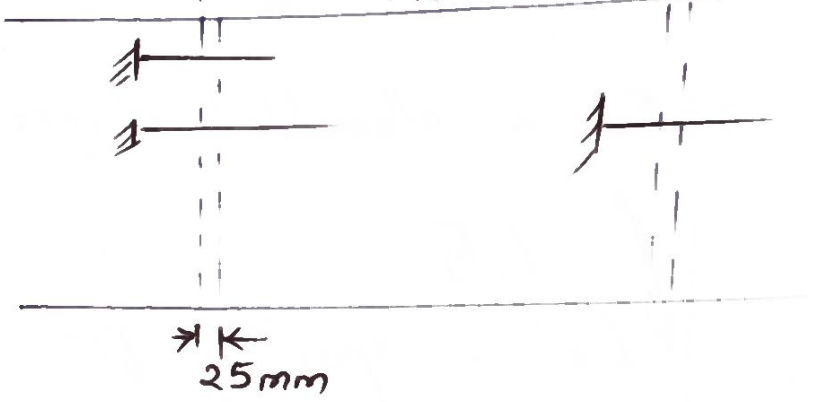
Expansion joint
- purpose
- thickness
- Gap.

Expansion joint

The purpose of expansion joint is to allow the expansion of the pavement due to rise in temperature wrt construction temperature.

Thickness : 25 mm to 2.5 cm

Gap : 140 m



- kaunsa bar provide kate h?
- kaise dete h dowel bars

Dowel bars are provided in expansion joints.

- Dowel bar is fixed @ one side and free at another side.

- Spacing formula

$$L_c = \frac{\delta}{\alpha [t_2 - t_1]}$$

δ is maximum expansion in slab length.

15.2

Contraction jt

- purpose

- Spacing kitne se zyada hai honi chahie

- (i) Pcc slab ki spacing ka formula

Contraction joint

The purpose of contraction joint is to allow the contraction of slab due to fall in temperature

distance / spacing ≯ 4.5m

(i) PCC Slab spacing

$$L_c = \left(\frac{2\sigma_c}{\gamma_c * f}\right)$$

σ_c is allowable stress in tension in c/c

$f = 1.5$

L_c = spacing bt^n contraction joints

- (ii) RCC slab ki spacing ka formula

(ii) for RCC slab.

$$L_c = \left(\frac{2\sigma_c}{\gamma_c * f}\right) * \frac{A_s}{Bh}$$

A_s is area of reinforcement

B is width of slab

h is thickness.

Marshall stability test.
↳ kiske lie krteh?

15.3

Marshall's stability test for mix design for flexible pavement

flowchart of marshalls stab test

Gt

↓

Gm

↓

% air voids V_v

↓

% volume of bitumen V_b

↓

VMA

↓

VFB

15.4

Marshall graphs

- kaunse do increase hote h phir decrease hote h?

- kaunsa decrease hota h

- kaunsa always ↑ ho rha h

- flow value wala kaisa h graph?

- VMA wala kaisa banta h?

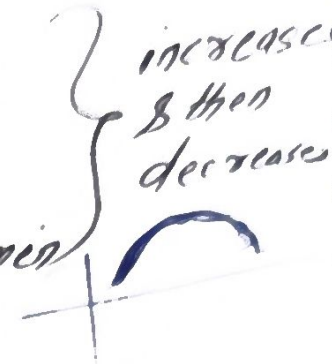
Marshalls graphs

① density vs ⅟.age bitumen

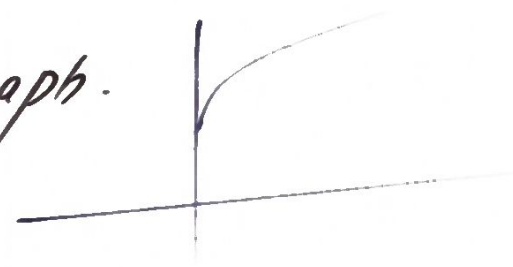
Marshalls stability number vs ⅟. bitumen — increases & then decreases

Marshall stab no ∝ Actual → maximum load value

② ⅟.age voids of air v/s ⅟. bitumen — always decreasing graph

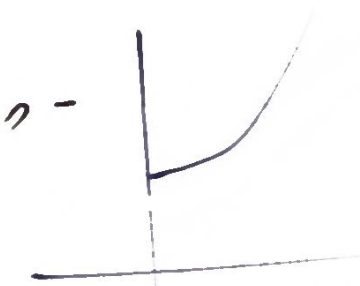


③ ⅟.age voids filled with bitumin (V_{FB}) — always increasing graph.



④ flow value vs ⅟. bitumen

Marshalls max
Flow value means deformation.
ka maximum deflection in mm.



⑤ VMA (voids in aggregate) vs ⅟. bitumen

VMA
$V_b + V_v$

Q * Imp one liners.

(Revelling failure occurs in bitumen pavement or flexible pavement is due to - - - -

→ open graded aggregate.

Q In hill roads the prevention of landslides for monsoon season is done by - - - - - -

→ 1) Retaining wall
2) Catch water drain
3) Breast wall

Q 1996 ES In bituminous construction compaction by pneumatic roller also is specified by - -

→ Pre mix carpet

Q Binders which is recommended in wet and cold climate - - - - -

→ cutback

JES 1998
Marshall design specifications
- MSN (m/n)
- flow value range
- % air void range
- VFB

Marshall design specification.

(i) Marshall stab no : $\overset{min}{340}$ kg

(ii) flow value [0.25mm unit] : 8 to 17

(iii) % air voids in mix : 3 to 5%.

(iv) Voids filled with bitumen = 75 - 85%.

15. (c) Q To carry out bituminous patchwork during rainy season most suitable binder is - - - - - - - - -

→ bituminous emulsion.

Q Bitumen grade 80/100 indicates penetration value = - mm to - mm

→ 8mm to 10mm

CC road construction steps kya kya h?

- prep
- place
- Mix
- Compact
- Cur

Steps involved in construction of CC roads

1. preparing the subgrade & base course & watering
2. placing the formwork / framework on the prepared base.
3. Mixing & placing of concrete
4. Compaction and floating.
5. Curing

(i) $\frac{80}{100}$ penetration → bituminous road

(ii) $\frac{85}{25}$ blown bitumen → mastic asphalt

(iii) MC - 70 cutback → prime coat.
medium curing

(iv) RT 5 → used in grouting
[Road tar 5]

QES 2009 Which property is not a desirable property of subgrade soil as a highway material. — — → (a) stability
(b) ease of compaction
(c) good drainage
(d) bitumen adhesion

A sequence in increasing order formance qualities

SD < PMC < SDBM
SD: wet per & casing Surface dressing
PMC: premix concrete
SDBM: semidense bituminous macadam
AC: asphaltic concrete
MA: mastic asphalt

Standard bitumen value as per marsh
11 = — — — — 6%

15.9

Railwayeng

- Gauge Defn
- Sizes in mm & Inches
  - BG
  - MG
  - NG
  - LG.
  - Std gauge
- Selection
  - coc
  - ~~t~~
  - Dslpmt
  - Phy feat
  - Spd.

Railway Engineering

1) Gauge of railway track is defined as clear distance between inner or running faces

(i) Broad gauge = 1.676m 5'6"

(ii) metre gauge = 1m 3 feet 3.3 inch

(iii) narrow gauge (NG) 0.762m 2'6"

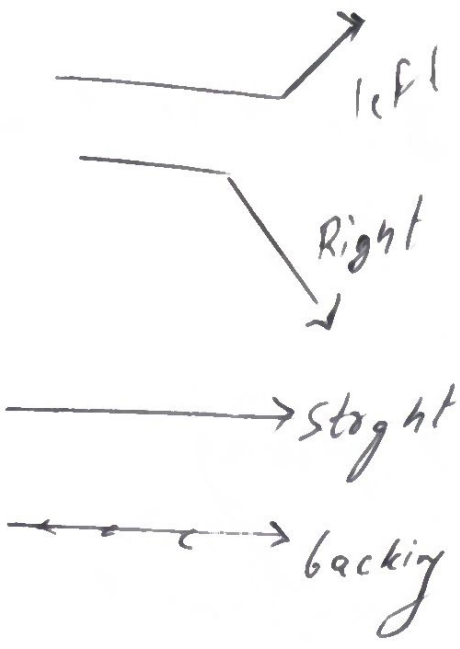
(iv) feather/feeder (LG) gauge = 0.610m 2 feet

(v) Standard gauge used in delhimetro = 1.435m

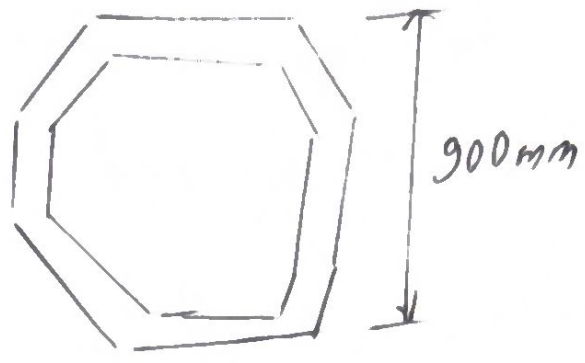
2) Selection of gauge depends on

1. cost of construction.
2. Volume & nature of traffic.
3. Developement of area.
4. Physical features of country
5. Speed of movement.

Vehicle mvmt



Traffic signs



Stop

Give way

Severity

PDO

Injury

fatal

Road surface

C Dry, Clear

W Wet

S Snowy icy

O other

lighting

D Day light

N Dark no lights

L Dark with Streetlights

Speed limit

Vehicle type

automobile

(T) truck

(B) Bus

(M) Motorcycle

(O) other

Pedestrian

uninvolved.

key to collision diagram symbols

Accident types

Rear end

Head on

L

sideswipe same direction

sideswipe opposite

out of control

collision with fixed object.

16.1

Class 16 23 Aug 2023

P-way

for railway or train an ideal P-way is required.

Requirements for an ideal pway

1 Requirement
Gauge

1. the gauge should be uniform & ideal.

b level

2. Both the rails should be @ same level in straight track.

c elevn

3. On curve proper superelevation should be provided to the outer rail.

d str

4. track should have enough lateral strength.

e city

5. track must have certain amount of elasticity.

f drng

6. drainage system should be perfect

g renewal
h joints

7. It should have adequate provision for easy renewal & repair

Coning of wheels

- kyo zaroorath
- level track pe wheel tread dia kaise vary karta h.
- circumference kaisa rehna chahiye str8 track pe?

& all joints / points & crossing (16.2
should be properly designed.

Coning of wheels

. it is required for the wheel to travel equal distances while travelling in a straight track.

. on the level track the axle may move towards one rail the diameter of wheel tread over the rail increase while it decreases over the other rail.

. on straight & level track the wheel remains in central position & the circumference travelled by both the wheels should be equal.

16.3)

coning of wheels

- Advantage wear & tear
  - it reduces wear & tear of wheel flanges
- advantage negotiate curve
  - it helps vehicle to negotiate curve smoothly.
- advantage central posn.
  - it keeps the train in central position
- Disadvantages
  - man diff
    - manufacturing difficulty
  - edge damage
    - no base plate are provided hence outer edge of rail gets damaged

Adzing of sleepers

- kya hota h adzing?

Adzing of sleepers

- A groove having an angle of 1 in 20 is being cut on the top of the sleeper so that the rail should be fixed in this location it is called adzing.

Canting of rail
- kyo krte h canting

Canting of rail

in order to minimise the dis advantage due to coning of wheels canting in rail is done. which means the rails are not placed flat but are tilted inward. (slope 1 in 20)

Rail
- kya hota h?

Rail

- Rails are member of railway track which are parallel to each other or parallel lines.
- It defines a track on which train travels.

- kyo krta h?

- Requirements
  - stl
  - wrng satc
  - foot thick

Requirement of good rails

1 proper composition of steel
2 deep & hard wearing surface of head
3 sufficient foot thickness & foot width

16.5

Rail

- type of rail sections
  - BH
  - DH
    - advantage
  - FF
- Rail sections used in india
  - for BG (wt & length & wt)
  - Metre gauge length & grades
  - narrow gauge grade & length

Type of rail sections

① bull headed rail (BH rail)

② Double headed rail (DH rail)

It can be reused after inverting.

③ flat footed rail. (FF rail)

Type of rail sections used in india.

for broad gauge → [IRS] 52 kg/m [13 m]

for metre gauge → 90 R [12 m rail length]
75 R
60 R

for narrow gauge → 50 R [12 m rail length]

Given sleeper density for broad gauge is n+6 if the track laid in 13 m rail length and length of welded rail is 26 m. find out number of sleeper reqd for railtrack length 1690m

Sleeper density calculn

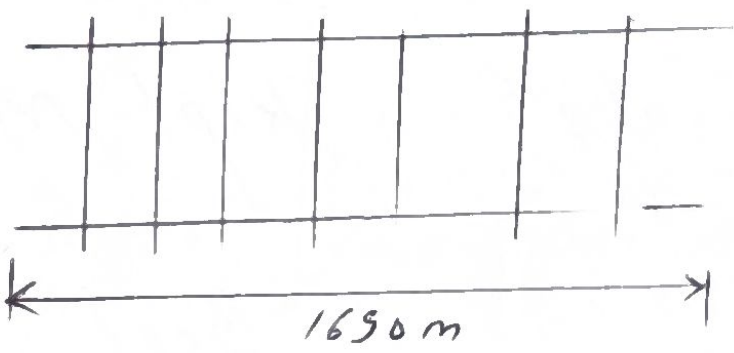
Ans: No sleepers reqd per rail = N+6 = 13+6
= 19 sleeper

* 1 rail length me sleeper reqd = 19 sleepers (13m)

* 1m length me sleeper reqd = 19/13 sleepers

∴ 1690m me sleeper reqd = $1690 \times \frac{19}{13}$

= 2470 sleepers



Types of sleeper

- W
- S
- CI
- CC

Types of sleeper

1. Wooden (Preferred on joints)
   - Hardwood
     - Sal t.
     - teak (Best)
   - Softwood
     - Chir
     - Deodar
2. Steel
3. Cast iron
4. Concrete.

about 20% tracks of indian railway are steel sleepers.

16.7 Wooden Sleeper

Sizes (BH

- BG
- MG
- NG

Sleeper density

- kaise check
- concrete?

Size of wooden sleeper

BG → 275 x 25 x 13 cm

MG → 180 x 20 x 11.5 cm

NG → 150 x 18 x 11.5 cm

Sleeper density as per Indian railways is normally kept M+2 to M+7.

Best wood for wooden sleeper → teak

Ballast

- kya hota hai

Ballast

- Big size aggregate, broken stones are laid & packed below and around sleeper is called ballast

Ballast

- functios
  - Distribute
  - elastic
  - drng
  - foundn
- Specifions
  - hrd
  - not wear
  - no organic

functions of ballast

1. Distribute the load from sleeper to the base or formation
2. Provides elasticity to the trail
3. Provides good drainage of the track.
4. Provides a suitable foundation for sleepers.

Specifications of good ballast

① it should be hard & durable

② it should not wear out due to physical reasons

③ it should be free from organic & inorganic residues.

16.2

Ballast → Type & Size of sleeper

Ballast sizes Type of sleeper

① 50 mm

② 40 mm

③ 25 m

Wooden sleeper, CI sleeper

CST 9 (central standard trial 9

point & crossing.

Ballast
- Specifics
  - a abrasion val
  - b impact val
  - c Spec grav
  - d Max wtr absptn
  - e elongn & flakiness index

Specification of good ballast

abraision val ≯ 30%.

impact val ≯ 20%.

Specific gravity ≯ 2.65

Max water absptn ≯ 10%.

elongn & flakiness index ≯ 50%.

Wheel se rail me load kaise jata hai

The transfer of wheel load from ~~train~~ wheel to rail is ~~called~~ in form of point load.

Class 17 24 Aug 24 [17.8.

Turnout ka section (point & crossing) ka mein diyah.

Geometric design

- Speed of train
- Martins formula (Safe speed of train)

on transition curve broad gauge & metre gauge = V = $4.35\sqrt{R-67}$

Velocity : $4.35\sqrt{\text{Radius in m} - 67}$

Maximum speed based on Railway board formula • speed on Transition curve

(i) for broad gauge = $V = 0.27\sqrt{(C_d + C_a)R}$

(ii) for metre gauge = $V = 0.347\sqrt{(d + C_a)R}$

(iii) for narrow gauge $V = 7.65\sqrt{R-6}$

Max speed upto 50 kmph.

17.2)

C_a is actual cant / superelevation

C_d is cant deficiency

V is in km/hr

R is in m.

⇌ speed or avg speed

- for $V_m > 50$ kmph

  $V_e = \frac{3}{4} V_m = 0.75 V_m$ } ↓

  $= V_{safe}$

- for $V_m < 50$ kmph

  $V_e = V_m$ } ↓

  $= V_{safe}$

Cant (Superelevation)

- It is the difference btn the height of outer rail to inner rail between on a curve.
- it is same as superelevation term

- Cant
  - inner rail is maintained @ original level & it is also called gradient rail.
- -ve superelevn

Transition curve

- ~~Types~~ Transition curve connects the straight section of train track @ one end and circular curve @ other end.
- Types
  1. Euler's spiral.
     - Ideal transition curve
  2. Cubic spiral
  3. Bernoullis lemniscate
  4. Cubic Parabola
     - Mostly used in indian railway
     - $y = \frac{x^3}{6RL}$
       - x & y are horiz & vert coordinates of length of x curve

Transn curve

- Types
  - Cubic parabola
    - shift: this is the amount by which a circular curve is shifted inwards so as to meet a transition curve
      - formula: $L^2/24R$
  - Extra widening
    - widening of gauges on curve
    - Reason - Centrifugal force
      - 2) Relative distance travelled by wheel
      - 3) Rigidity of wheel base.
    - $We = W = \frac{13(B+L)^2}{R}$
      - B is wheel base
        - BG: 6m
        - MG: 4.88m
      - L is Lap of flange
        - $0.02\sqrt{h^2 + 4Dh} = L$

rawide

ing

We : $\frac{13\ (B+L)^2}{R}$

- h is depth of flange below top of rail in cm
- D is diameter of wheel in cm.

transition curve

$\frac{4}{5}$ of Calculated speed.

of setting-out horizontal curve in field

dolite method

ential offset method

d deflection method

chord offset method

17.6

Grade compensn

Broadgauge	0.04% /°C	or 70/R
Metre gauge	0.03½ °C	or 52.5/R
Narrowgauge	0.02½ °C	or 35/R

R is radius of curve in m.

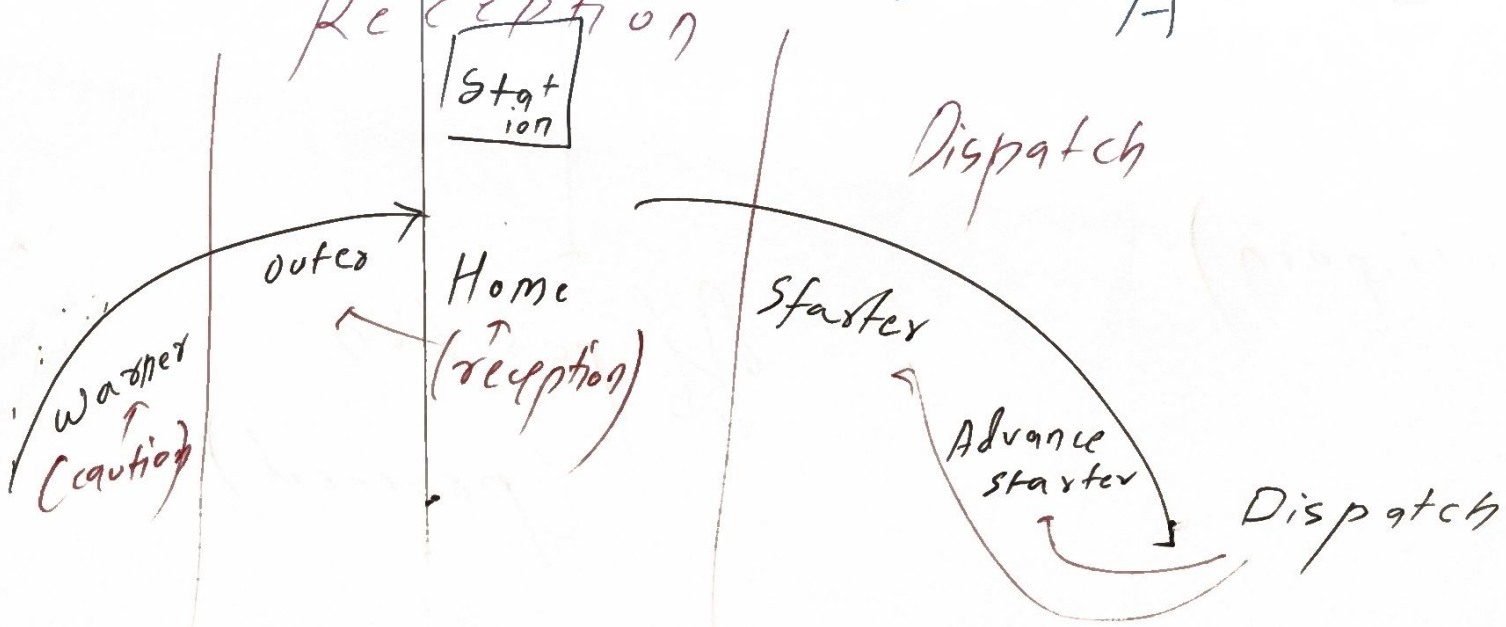
Signal

Signal Sequence

#2 times pucha h.

W - O - H - S - A

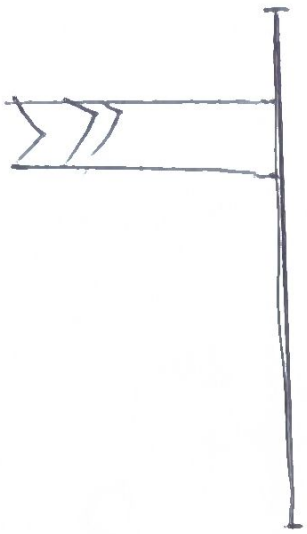
Reception



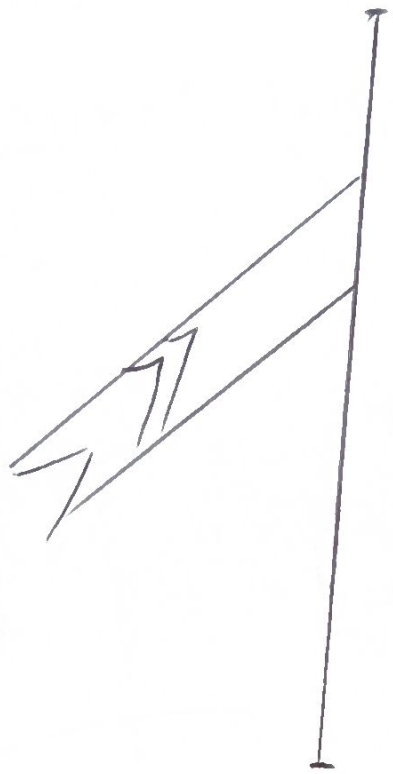
formation : gaige + width in railway shoulder + drain.

flowchart →

10.2



Stop
(ON poin)



Off position
(proceed)

Disk signal
alternate name → shunting signal — used in station yard for shunting operation

a Mechanical devices used for interlocking the track or interlocking →

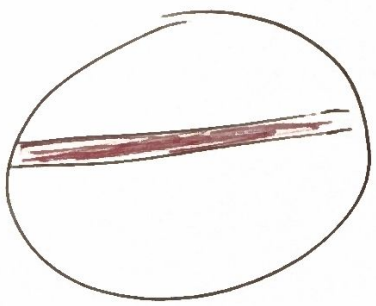
(1) point lock
(2) Toad/er bar
(3) Detector
(4) Connecting devices.

Semaphore Signal → These are movable arm signals

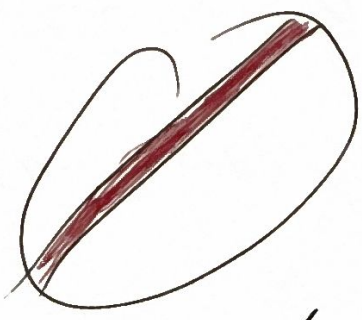
Warner signals → Defines that the train has to move @ a slower speed @ certain locations.

fixed signals are most of the time placed @ the top of the post.

Shunting signal

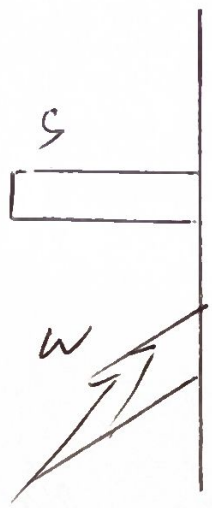


Stop

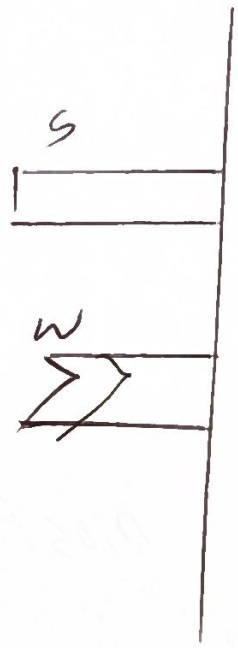


proceed

18·5) Q When semaphore & warner signals are installed in same post then for stop indicn position - both arms should be horizontal.



proceed



stop

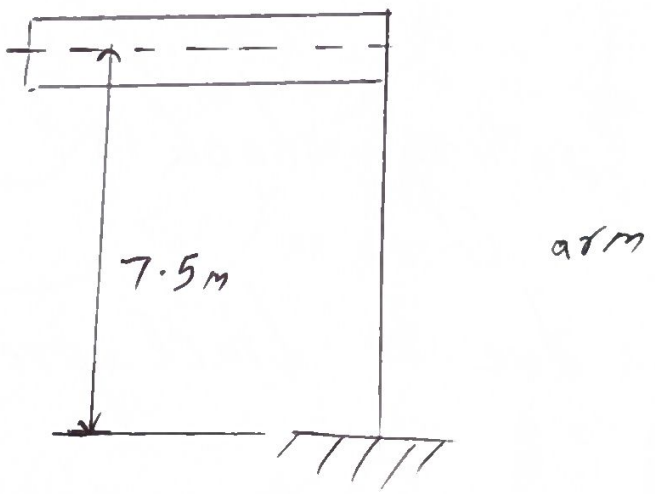
Q Yellow light hand signal → prepare or proceed cautiously.

Q The reception signals are

1) Outer

2) Home signal

Q The height of centre of arm of semaphore signal above the ground is 7.5 m.



Q The treadle bar is provided - near and parallel to inner side of one of the rails.

Q The object of providing point lock is - to ensure that each switch is correctly set.

Q Lock bar - it is a mechanical device which is used to ensure that root cannot be changed while the train is on the point even after putting back the signal.

18.7 Track resistance and power of locomotive

1) Resistance to traction.

A) Resistance due to rolling stock — Indep. of speed
B) Resistance due to track profile — dep. on speed
C) Resistance due to tractive effort — atmospheric resistance
D) Resistance due to atmospheric / climatic condition.

A) Resistance due to rolling stock.

→ Resistance which is independent of speed

formula: 0.016 W
W → weight of train in tonnes.

2 → Resistance dependent on speed

$RT_2 = 0.0000,8\ wV$

↳ speed of train in kmph

↓ wt in tonne

3 → atmospheric RT

$RT_3 = 0.000,000,6\ wV^2$

Train resistance $RT = RT_1 + RT_2 + RT_3$

Hauling capacity of locomotive:

It can be defined as the total load that can be handled by the locomotive

It indicates is an indicative of power available

16.9

HC > TR train resistance
/ traction resistance

Hauling capacity formula.

HC = μ W
↳ wn

W = total load on driving wheel

w = weight on driving axle..

n = no of pairs of driving wheels

μ = coeff of friction

Tractive effort formula.

It is taken equal to or little greater than hauling capacity.

HC > T resistance

HC ≤ Trac. effort

Tractive effort : $T_e = \frac{P d^2 L}{D}$

→ for steam locomotive

→ diesel locomotive

$$T_e = \frac{308 (HP_f)}{V}$$

HP_f → Rated horsepower of engine

V → kmph

Tractive effort for steam loco depends on

1) Difference in pressure of cylinder (P)
2) length of stroke (L)
3) Diameter of piston (d)
4) Area of piston (A)
5) Diameter of wheel (D)